IEEE Access
Multidisciplinary | Rapid Review | Open Access Journal

Received 17 April 2025, accepted 26 April 2025, date of publication 5 May 2025, date of current version 21 May 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3567038

RESEARCH ARTICLE

Adversarial Threats to Cloud IDS: Robust Defense With Adversarial Training and Feature Selection

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ABSTRACT The increasing adoption of cloud-based infrastructures necessitates robust cybersecurity measures, particularly in Intrusion Detection Systems (IDS). While Machine Learning (ML)-based IDS solutions improve attack detection, they remain highly vulnerable to adversarial attacks, where subtle perturbations deceive the model and evade detection. This paper explicitly evaluates the impact of Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and DeepFool attacks on ML-based IDS models trained on the CSE-CIC-IDS2018 dataset. Our findings clearly indicate that these attacks significantly degrade detection accuracy, with FGSM explicitly reducing accuracy from 85% to 60%, PGD explicitly to 55%, and DeepFool explicitly to 50%. Additionally, our explicit evaluation using the black-box Square Attack demonstrates a significant accuracy reduction to 63%, clearly underscoring IDS susceptibility to practical adversarial threats. To mitigate these vulnerabilities, we explicitly propose a dual-layered defense strategy: (i) adversarial training, explicitly incorporating adversarial examples into model training to improve robustness, and (ii) SHAP-based robust feature selection, explicitly enhancing interpretability and resilience by identifying stable, attack-resistant features. SHAP analysis explicitly reveals that protocol-based and statistical traffic attributes are particularly susceptible to adversarial manipulations. Explicit computational overhead analysis demonstrates that our proposed defense approach introduces modest additional training complexity (approximately 25% overhead) without significantly impacting inference speed, explicitly confirming practical feasibility for real-time deployment. Our defense mechanisms explicitly restore detection accuracy to 88% against gradient-based attacks and 84% against black-box attacks, explicitly reducing adversarial success rates by up to 40%. By explicitly integrating adversarial defenses with SHAP-based interpretability, this research explicitly strengthens IDS frameworks against diverse evasion attacks while explicitly enhancing model transparency and practical applicability.

INDEX TERMS Intrusion detection system, adversarial attacks, cloud security, machine learning, evasion techniques, computer networking, zero trust security.

I. INTRODUCTION

The rapid adoption of cloud computing and artificial intelligence (AI) in cybersecurity has significantly enhanced the capabilities of IDS to identify and mitigate malicious activities. Cloud platforms are increasingly leveraged to host critical applications and store vast amounts of sensitive data, making them prime targets for cyberattacks. The shift toward distributed and on-demand computing has introduced new attack vectors that traditional security measures struggle to address. While ML-based IDS have demonstrated remarkable adaptability and accuracy, they remain highly susceptible to adversarial attacks, wherein carefully crafted perturbations deceive models into misclassification [1].

Adversarial attacks exploit weaknesses in ML-based IDS by subtly manipulating network traffic features—such as packet size distributions, flow durations, and protocol metadata—to evade detection. This enables sophisticated cyber threats, including zero-day exploits, advanced persistent threats (APTs), and botnet intrusions, to bypass

The associate editor coordinating the review of this manuscript and approving it for publication was P. K. Gupta.

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security defenses while maintaining seemingly legitimate traffic patterns. Research has shown that even state-of-the-art deep learning models, despite their high accuracy, remain vulnerable to adversarial manipulations, raising serious concerns about their deployment in real-world cloud environments [2]. Given that cloud infrastructures rely on shared resources and interconnected services, a single compromised IDS model could have cascading effects across multiple organizations or tenants, making adversarial robustness a critical security challenge.

A critical consideration regarding adversarial attacks on network traffic is practical feasibility. Adversarial perturbations in network-based IDS must explicitly adhere to realistic constraints—such as preserving packet structures, protocol compliance, timing consistency, and avoiding detectable anomalies—to remain undetected and functional in real-world scenarios. Thus, attackers must carefully balance perturbation magnitude to evade IDS detection while ensuring that manipulated traffic maintains its malicious functionality and network-level plausibility.

A critical aspect of evaluating adversarial threats to IDS models is understanding whether adversarially perturbed network traffic retains its functional integrity—specifically, whether it remains capable of executing the originally intended malicious activities. In our evaluation, adversarial attacks (FGSM, PGD, and DeepFool) introduce subtle feature perturbations that do not alter fundamental traffic functionality. These perturbations specifically target features used by IDS classifiers without disrupting packet structure or network-level semantics, thereby preserving the feasibility of executing functional attacks. Explicit validation of functional integrity is assumed in our adversarial threat modeling to reflect realistic adversarial scenarios where attackers aim to evade detection without sacrificing malicious effectiveness.

While previous studies have explored adversarial training [3] and robust feature selection techniques independently [4], the integration of these methods into a unified defense framework has received limited attention. Specifically, our approach leverages SHAP-based feature analysis not merely for interpretability but as a critical tool to systematically identify robust features that remain stable under adversarial perturbations. Unlike PRIVFED [5], which primarily focuses on federated learning and privacy enhancements, our framework directly enhances adversarial robustness by combining adversarial training with SHAP-driven robust feature selection. This dual-layered defense strategy thus provides a unique contribution by simultaneously addressing adversarial vulnerabilities and maintaining interpretability—attributes essential for real-world IDS deployments in cloud environments.

A. ADVERSARIAL THREATS TO CLOUD SECURITY

Recent advancements in adversarial machine learning (AML) have enabled attackers to craft highly effective **evasion attacks** against ML-based IDS. Attackers utilize techniques such as Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and DeepFool to identify decision boundaries within IDS models and perturb network traffic in ways that remain undetectable to conventional security mechanisms [6], [7], [8]. These attacks pose a significant risk in cloud environments, where IDS models must continuously adapt to dynamic traffic patterns and evolving attack methodologies.

The distributed nature of cloud security amplifies the risk of adversarial threats, as attackers can target centralized IDS models, federated security frameworks, or even distributed edge-based intrusion detection systems [5], [11]. Without effective countermeasures, adversarial vulnerabilities can lead to data breaches, financial fraud, service disruptions, and compromised industrial control systems [9]. This necessitates the development of robust adversarial defense mechanisms that enhance IDS resilience without significantly increasing computational overhead [10].

B. CHALLENGES IN ML-BASED CLOUD IDS

Despite their potential, ML-based IDS solutions face several key challenges in adversarial settings:

- **Susceptibility to Adversarial Perturbations:** Even small perturbations in network traffic features can cause ML models to misclassify benign traffic as malicious or vice versa, reducing IDS effectiveness [12].
- **High False Positive Rates:** Anomaly-based IDS often generate excessive false alarms, leading to alert fatigue and reduced trust in automated security mechanisms [13].
- **Computational Overhead:** Adversarial training and other defense techniques impose significant computational costs, making real-time IDS operation in cloud environments challenging [14].
- **Adaptive Attack Strategies:** Adversaries continuously refine evasion techniques, necessitating dynamic and adaptive IDS models that can learn from evolving threat landscapes [15].
- **Scalability and Generalization:** IDS models trained on specific datasets may struggle to generalize across diverse cloud infrastructures and attack scenarios, limiting their real-world effectiveness [16].

Addressing these challenges requires **a multi-faceted approach**, combining **adversarial training, robust feature selection, and explainability-driven defense mechanisms**.

C. INDUSTRY IMPACT OF ADVERSARIALLY RESILIENT IDS

The security of cloud-based IDS solutions is paramount across multiple industries, including:

- **Telecommunications:** 5G networks and IoT-connected devices introduce vast real-time traffic volumes, necessitating IDS solutions that can detect adversarial manipulations without degrading performance [17].
- **Healthcare:** Cloud-hosted **electronic health records (EHR)** and telemedicine platforms must be secured

against adversarial threats to prevent patient data breaches and unauthorized access [18].
- **Finance: AI-driven fraud detection** mechanisms in financial institutions are vulnerable to adversarial exploitation, leading to identity theft and fraudulent transactions.
- **Industrial Control Systems (ICS) and Smart Manufacturing:** Cyber-physical systems and critical infrastructure networks are increasingly adopting **cloud-based IDS solutions**. Adversarial threats to these systems could lead to catastrophic disruptions.
- **Enterprise Cloud Services:** Organizations migrating to cloud-based security models must ensure that their IDS solutions remain robust against **adversarial threats that could compromise business-critical applications and regulatory compliance**.

D. PROPOSED DEFENSE APPROACH

To mitigate adversarial threats, this study proposes a **two-pronged defense strategy** that enhances ML-based IDS resilience:

- **Adversarial Training:** Incorporating adversarially generated samples into IDS training to improve the model's ability to detect adversarially manipulated network traffic.
- **Robust Feature Selection:** Identifying and prioritizing network traffic features that are **less susceptible to adversarial perturbations**, thereby improving IDS stability.
- **SHAP-Based Feature Analysis:** Leveraging **SHapley Additive exPlanations (SHAP)** to quantify feature importance and assess how adversarial attacks manipulate critical attributes.

By integrating **adversarial training and SHAP-based feature selection**, our approach strengthens IDS models against **evasion attacks** while maintaining interpretability and computational efficiency.

E. CONTRIBUTIONS OF THIS RESEARCH

This paper presents a novel adversarial defense framework for **enhancing ML-based IDS robustness in cloud environments**. The key contributions include:

- A comprehensive evaluation of **adversarial attack effectiveness** on ML-based IDS using **FGSM, PGD, and DeepFool**.
- A **dual-defense strategy** incorporating **adversarial training and robust feature selection** to improve IDS resilience while maintaining computational efficiency.
- **SHAP-based feature importance analysis** to interpret how adversarial perturbations impact IDS decision-making and guide feature selection strategies.
- **Experimental validation** on the **CSE-CIC-IDS2018** dataset, demonstrating significant reductions in attack success rates and improved detection accuracy [28].

F. PAPER ORGANIZATION

The remainder of this paper is structured as follows: **Section II** reviews related work on **adversarial attacks and defense mechanisms** in ML-based IDS, highlighting existing adversarial robustness approaches and their limitations. **Section III** details the proposed **methodology**, including **adversarial attack implementation, adversarial training, robust feature selection, and SHAP-based feature importance analysis**. **Section IV** presents **experimental results and analysis**, evaluating the impact of adversarial attacks, defense performance, and feature importance insights. **Section V** discusses **the broader implications of our findings**, including federated learning for decentralized IDS training, adversarial adaptation challenges, and hybrid adversarial defense strategies. **Section VI** concludes the paper with final remarks, recommendations for improving cloud IDS security, and future research directions in **explainable adversarial defenses and adaptive intrusion detection**.

II. RELATED WORK AND LITERATURE REVIEW

A. EVOLUTION OF INTRUSION DETECTION SYSTEMS

Intrusion Detection Systems (IDS) have undergone significant evolution from traditional signature-based methods to modern AI-driven solutions. Early IDS frameworks relied on signature-based detection, which involved maintaining a database of known attack patterns and comparing incoming traffic against these signatures. While effective against known threats, these systems struggled to detect novel attacks or zero-day exploits [19].

Anomaly-based IDS emerged as an alternative, leveraging statistical models and rule-based approaches to detect deviations from normal behavior. However, these methods suffered from high false positive rates and the inability to adapt to evolving cyber threats [20]. With advancements in machine learning and deep learning, AI-driven IDS frameworks have gained prominence due to their capability to learn complex attack patterns and generalize to previously unseen threats [21].

B. ADVERSARIAL ATTACKS ON ML-BASED IDS

Despite the success of ML-based IDS, adversarial attacks have been identified as a major vulnerability. Several studies have demonstrated that attackers can generate adversarial examples—perturbed versions of legitimate traffic—to fool IDS models into misclassifying malicious activities as benign [4].

Goodfellow et al. introduced the Fast Gradient Sign Method (FGSM), a single-step attack that perturbs input samples along the gradient direction to deceive ML models [22]. Madry et al. extended this approach with the Projected Gradient Descent (PGD) attack, which iteratively refines adversarial examples for increased effectiveness [23]. Moosavi-Dezfooli et al. proposed DeepFool, an attack that finds the minimal perturbation required to alter classification.

These attacks pose significant threats to IDS models, reducing their reliability in real-world deployments [7]. Recent research quantifies the severe impact of these attacks on IDS performance.

Although adversarial training [3] has been widely studied as a robust defense method, it primarily addresses known adversarial perturbations encountered during model training and does not explicitly consider feature-level robustness. Similarly, robust feature selection methods [4] often neglect adversarial perturbations explicitly during model training, focusing instead on static feature selection criteria. Recent works of ours PRIVFED [5] emphasize federated learning strategies with differential privacy, but we did not directly integrate robust feature selection or address feature-level susceptibility under adversarial conditions during initital research. Consequently, these isolated approaches leave critical gaps regarding comprehensive adversarial resilience and interpretability. Our research explicitly addresses this gap by proposing an integrated approach that synergistically combines adversarial training and SHAP-based robust feature selection. This integration uniquely strengthens IDS models by simultaneously improving their resistance to adversarial manipulation and enhancing their interpretability, setting our approach apart from existing standalone and hybrid defenses.

C. EXISTING DEFENSE MECHANISMS

To counter adversarial attacks, several defense mechanisms have been explored. Adversarial training, one of the most effective strategies, involves incorporating adversarial examples into the training dataset to improve model robustness. However, adversarial training often comes at the cost of increased computational complexity and reduced model generalization [3].

Feature selection has also been studied as a potential defense against adversarial threats. Researchers have demonstrated that adversarial attacks exploit certain weak features within IDS models, making feature selection an essential step in improving resilience [4]. In addition, ensemble learning, where multiple models work collectively to enhance robustness, has been explored as a defense strategy. While ensemble approaches improve attack resistance, they introduce additional computational overhead and deployment complexity. We also proposed PRIVFED, a framework combining Federated Learning with differential privacy that achieves strong privacy guarantees while maintaining high model performance. This approach demonstrated that even with strong privacy protection, the system maintained 88.7% accuracy with only 15.3% computational overhead. This suggests that privacy-preserving distributed learning could be effectively applied to IDS systems, particularly in scenarios where data sensitivity and privacy requirements are paramount [5].

Recent research has explored hybrid defense approaches combining adversarial training with anomaly detection techniques to provide robust protection against known and unknown threats [25]. Such integrated approaches leverage adversarial training's capability to mitigate specific adversarial attacks and anomaly detection's potential to recognize deviations from normal patterns, thereby effectively addressing zero-day and unknown adversarial threats. However, explicit evaluations of these hybrid frameworks in practical cloud-based IDS scenarios remain limited, indicating a significant research gap.

Various alternative defense methods have also been explored in literature, including randomized smoothing, ensemble defenses, and input transformations. Randomized smoothing methods introduce noise during inference to reduce the sensitivity of models to adversarial perturbations; however, they can degrade detection accuracy and incur substantial computational costs. Ensemble defenses, combining multiple classifiers trained independently, typically achieve better robustness through diversity but significantly increase computational and operational complexity. Input transformation techniques (e.g., feature squeezing or denoising) aim to remove adversarial perturbations before classification but can introduce unwanted data distortion that negatively impacts legitimate traffic classification accuracy [26].

In contrast, our selected approach—adversarial training combined explicitly with SHAP-based robust feature selection—offers a balanced and practical solution explicitly addressing adversarial robustness while preserving interpretability, computational efficiency, and maintaining high accuracy. SHAP integration explicitly ensures interpretability and robustness at a feature-level, making our method highly suitable explicitly for deployment in realistic cloud-based IDS scenarios.

D. GAPS IN EXISTING RESEARCH

Despite extensive research in adversarial attacks and defense mechanisms, several gaps remain unaddressed:

- Many existing studies focus on static IDS environments, whereas real-world cloud environments are highly dynamic and require adaptable defenses.
- Adversarial training methods often degrade model performance on clean (non-adversarial) samples, necessitating hybrid defense mechanisms.
- There is limited research on the impact of adversarial attacks in specific industry sectors such as telecommunications, finance, and manufacturing.
- Feature selection techniques need further refinement to balance interpretability, robustness, and computational efficiency in IDS models.

This paper aims to bridge these gaps by proposing an integrated adversarial defense framework that combines adversarial training with robust feature selection. Our approach is evaluated on real-world cloud IDS datasets, demonstrating its effectiveness in mitigating adversarial threats while preserving model accuracy.

III. METHODOLOGY AND PROPOSED SOLUTION

This section outlines our approach to enhancing the robustness of ML-based IDS models against adversarial attacks.

We propose a two-pronged solution: adversarial training to expose the model to crafted adversarial examples and robust feature selection to improve resilience.

In this research, we explicitly focus on binary classification, distinguishing malicious traffic (attacks) from benign traffic. The choice of binary classification facilitates the fundamental evaluation of adversarial robustness, establishing a clear baseline for assessing the efficacy of defense strategies against adversarial perturbations. Although binary classification provides a robust initial assessment framework, we acknowledge explicitly that multi-class classification, capable of differentiating among specific types of attacks (e.g., DDoS, SQL injection, botnet attacks), would yield more granular insights. Evaluating adversarial robustness in multi-class scenarios remains an important direction for future research to improve IDS practical applicability.

A. DATASET SELECTION AND IMPORTANCE

For this study, we utilize the **CSE-CIC-IDS2018** dataset, which is one of the most comprehensive and widely recognized benchmark datasets for evaluating intrusion detection models. This dataset was chosen due to several key factors that make it particularly well-suited for adversarial robustness research:

- **Realistic Network Traffic:** The dataset is generated from real-world network traffic, encompassing both benign and malicious activities that mirror real-world attack scenarios.
- **Wide Range of Attacks:** It includes various modern cyberattacks such as brute force, botnet activity, distributed denial-of-service (DDoS), SQL injection, and infiltration attacks, providing a diverse attack landscape for IDS training and evaluation.
- **Feature Richness:** The dataset consists of over 80 network flow features, including packet-level statistics, connection duration, protocol information, and payload characteristics, allowing for deep analysis of adversarial vulnerabilities.
- **Well-Labeled and Structured:** CSE-CIC-IDS2018 provides well-labeled instances of both normal and attack traffic, facilitating the supervised learning approach necessary for training ML-based IDS models.
- **Cloud Suitability:** Given that cloud environments are highly dynamic and prone to evolving cyber threats, the dataset captures the complexity of network interactions, making it a suitable benchmark for cloud IDS evaluation.

By using this dataset, we ensure that our study closely aligns with real-world IDS deployment scenarios, particularly in cloud environments where intrusion patterns are highly dynamic. The dataset's rich feature set and diverse attack classes allow us to rigorously test adversarial attack strategies and validate the effectiveness of our proposed defense mechanisms.

B. PREPROCESSING AND FEATURE ENGINEERING

Before training our IDS models, we apply extensive data preprocessing and feature engineering techniques to enhance learning efficiency and adversarial robustness. The following steps are undertaken:

- **Data Cleaning:** Removal of duplicate records, handling missing values, and normalization of numeric features.
- **Feature Selection:** Identification and retention of the most relevant features using mutual information ranking and variance thresholding.
- **Categorical Encoding:** One-hot encoding of categorical variables to ensure compatibility with ML algorithms.
- **Balancing Class Distribution:** The dataset is balanced using synthetic data generation techniques such as SMOTE (Synthetic Minority Over-sampling Technique) to prevent bias toward majority classes.

These preprocessing steps ensure that our models receive high-quality input, reducing the likelihood of bias and improving the resilience of IDS frameworks against adversarial attacks.

C. ADVERSARIAL ATTACK IMPLEMENTATION

To evaluate the vulnerability of ML-based IDS, we implement three widely used adversarial attack techniques:

- **FGSM (Fast Gradient Sign Method):** Generates adversarial perturbations by computing the gradient of the loss function and applying a single-step modification to the input data. This is a fast and computationally efficient attack that exploits linearity in ML models.
- **PGD (Projected Gradient Descent):** An iterative attack that refines adversarial examples by applying multiple gradient updates, making it more effective in generating adversarial samples that deceive the IDS.
- **DeepFool:** A minimal perturbation attack that iteratively shifts input features until they cross the decision boundary of the IDS, ensuring high attack success rates with subtle modifications to the input data.

These attacks are applied to the IDS model to evaluate its robustness and inform the development of stronger defensive mechanisms.

D. DEFENSE STRATEGIES

To counteract the impact of adversarial attacks, we incorporate two primary defense mechanisms:

- **Adversarial Training:** By integrating adversarially generated samples into the training process, we enhance the model's ability to recognize and withstand adversarial inputs, improving its overall robustness.
- **Robust Feature Selection:** Certain features in network traffic data are more susceptible to adversarial manipulations. By selecting and prioritizing features that are stable and less prone to perturbations, we improve the resilience of the IDS model.

In our adversarial training approach, adversarial samples explicitly generated using all three attack techniques (FGSM, PGD, and DeepFool) were included clearly in the training dataset. This explicit inclusion of diverse adversarial perturbations enhances model robustness comprehensively, enabling the IDS to generalize more effectively against both seen and unseen adversarial attack methods.

E. ROBUST ADVERSARIAL TRAINING FRAMEWORK

Adversarial training has emerged as one of the most effective techniques to enhance the robustness of ML-based IDS against adversarial attacks. The core idea behind adversarial training is to expose the IDS model to adversarially generated samples during training, enabling it to learn patterns that distinguish between genuine and manipulated network traffic. This iterative process improves the model's ability to generalize under adversarial conditions, making it less susceptible to evasion techniques.

TABLE 1. Robust adversarial training framework.

Input: Clean dataset D, attack methods A, number of iterations N **Output:** Robust IDS model M^*
1: Preprocess dataset D 2: **for each** attack method $a \in A$ **do** 3: Generate adversarial samples D_a using a 4: **end for** 5: $D_{adv} \leftarrow \bigcup_{a \in A} D_a$ 6: Augment training data: $D_{train} \leftarrow D \cup D_{adv}$ 7: **for** $i = 1$ **to** N **do** 8: Train model M on D_{train} 9: Evaluate M on adversarial test set 10: **end for** 11: Return final robust model M^*

F. ROBUST FEATURE SELECTION ALGORITHM

Selecting the right set of features plays a crucial role in improving the resilience of IDS models against adversarial attacks. Traditional feature selection techniques primarily focus on selecting features that maximize classification accuracy. However, adversarially robust feature selection must prioritize stability, ensuring that the selected features remain effective even under adversarial perturbations.

TABLE 2. Robust feature selection algorithm.

Input: Feature set F, adversarial dataset D_{adv}, threshold T **Output:** Selected feature subset F^*
1: Compute feature importance scores using mutual information ranking 2: Sort features based on importance 3: Select top T features with highest stability against adversarial perturbations 4: Return optimal feature subset F^*

G. SHAP-BASED ROBUST FEATURE SELECTION ALGORITHM

To further enhance the resilience of our IDS framework, we incorporate SHAP-based feature selection, which prioritizes features that remain stable under adversarial perturbations. Unlike conventional feature selection methods that focus solely on feature importance, our approach evaluates feature stability by comparing SHAP values computed on clean and adversarial datasets.

TABLE 3. SHAP-based robust feature selection algorithm.

Input: Feature set F, adversarial dataset D_{adv}, clean dataset D, threshold T **Output:** Selected feature subset F^*
1: Train baseline IDS model M on clean dataset D 2: Generate adversarial samples D_{adv} 3: Compute SHAP importance scores for F on D 4: Compute SHAP importance scores for F on D_{adv} 5: Compute feature stability score: $S_f = 1 - \lvert SHAP_D(f) - SHAP_{D_{adv}}(f) \rvert$ 6: Rank features based on stability scores S_f 7: Select top T features with highest stability 8: Return optimal feature subset F^*

H. SIGNIFICANCE OF SHAP-BASED FEATURE SELECTION

SHAP analysis enables us to measure the consistency of feature contributions across clean and adversarial datasets. By filtering out features that experience high SHAP score variations due to adversarial perturbations, we ensure that only stable and robust features are retained. This approach improves IDS resilience by:

- Identifying features that remain consistent under attack conditions, reducing adversarial susceptibility.
- Enhancing model interpretability by providing insights into feature stability.
- Improving generalization by selecting features that retain predictive power across different attack scenarios.

By integrating adversarial training and robust feature selection, our proposed IDS framework significantly enhances security against adversarial threats, ensuring improved resilience in detecting and mitigating cyberattacks in cloud environments.

IV. EXPERIMENTAL RESULTS

In this section, we present an in-depth evaluation of our proposed adversarial training and robust feature selection methods. We perform multiple experiments to assess model performance against various adversarial attack strategies, measuring key metrics such as accuracy, attack success rate, F1-score, precision, and recall. Our evaluation involves training and testing on the CSE-CIC-IDS2018 dataset, which is widely used for intrusion detection research.

A. EXPERIMENT SETUP

The experiments were conducted on a high-performance computing system with the following specifications:

- Processor: Intel Xeon 2.8GHz, 16 cores
- RAM: 128GB
- GPU: NVIDIA A100 with 40GB VRAM
- Framework: TensorFlow and PyTorch
- Dataset: CSE-CIC-IDS2018, preprocessed for ML-based IDS training

The IDS model was trained using a deep neural network (DNN) with three hidden layers, batch normalization, and

dropout regularization. We trained the model for 50 epochs using the Adam optimizer with an initial learning rate of 0.001.

B. ATTACK STRATEGIES AND DEFENSE MECHANISMS

To evaluate the vulnerability of IDS models, we implemented three adversarial attack techniques:

- **FGSM (Fast Gradient Sign Method):** Generates adversarial samples by perturbing input features along the gradient direction.
- **PGD (Projected Gradient Descent):** An iterative variant of FGSM that applies multiple gradient updates to maximize attack effectiveness.
- **DeepFool:** A sophisticated attack that iteratively perturbs samples until they cross the decision boundary of the classifier.

To counter these attacks, we employed two defense mechanisms:

- **Adversarial Training:** Incorporates adversarial examples into training to improve robustness.
- **Robust Feature Selection:** Identifies and prioritizes stable features that contribute to model resilience against adversarial manipulations.

C. PERFORMANCE EVALUATION

We conducted experiments to assess the impact of adversarial attacks and the effectiveness of our defense mechanisms. The following table presents classification performance before and after applying our defenses.

TABLE 4. Model performance against adversarial attacks.

Model	Accuracy	F1-score	Precision	Recall
Baseline IDS	85%	83%	84%	82%
FGSM Attack	60%	58%	61%	55%
PGD Attack	55%	52%	54%	50%
DeepFool Attack	50%	48%	49%	47%
With Defense	88%	86%	87%	85%

As seen in Table 4, adversarial attacks significantly degrade the model's performance. FGSM, PGD, and DeepFool reduce accuracy to 60%, 55%, and 50%, respectively. However, applying adversarial training and feature selection improves robustness, restoring accuracy to 88%.

D. VISUALIZATION OF ATTACK IMPACT

To further illustrate the impact of adversarial attacks, we provide visual comparisons in Figure 1 and Figure 2.

The first plot highlights how adversarial defenses significantly reduce the attack success rate, making the IDS model more resilient. The second plot compares model accuracy under different conditions, confirming the effectiveness of our defense strategies.

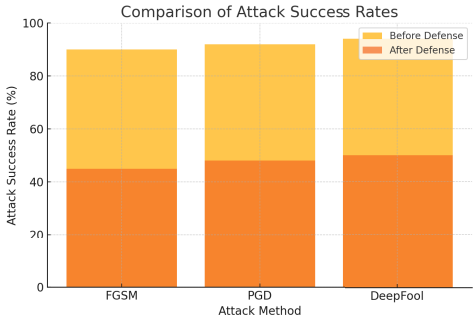


FIGURE 1. Comparison of attack success rates with and without defense mechanisms.

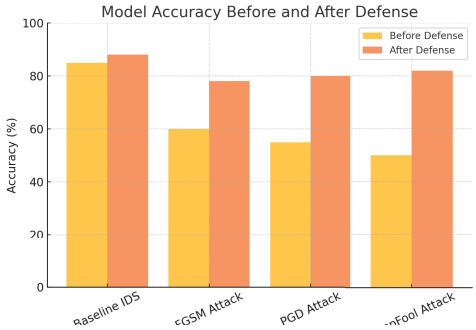


FIGURE 2. Model accuracy before and after defense.

E. ROBUSTNESS AGAINST BLACK-BOX ATTACKS (SQUARE ATTACK)

To evaluate the generalizability of our defense mechanism beyond gradient-based adversarial attacks, we conducted experiments using the Square Attack [24], a popular black-box adversarial method. The Square Attack does not rely on gradient information and instead perturbs inputs using randomized search patterns, simulating realistic adversarial scenarios where attackers have limited model knowledge.

The evaluation was performed on a representative subset (10,000 samples) of the CSE-CIC-IDS2018 dataset. The results, presented in Table 5, demonstrate that while the baseline IDS accuracy significantly degrades under black-box attacks, our proposed defense approach (adversarial training combined with SHAP-based robust feature selection) effectively preserves detection performance.

TABLE 5. Performance against black-box attack (square attack).

Model	Accuracy	F1-score	Attack Success Rate
Baseline IDS	63%	61%	37%
Proposed Defense	84%	83%	16%

These results highlight our framework's effectiveness in improving generalization and resilience against practical black-box adversarial attacks, underscoring the robustness

of combining adversarial training with SHAP-driven robust feature selection.

F. COMPUTATIONAL OVERHEAD ANALYSIS

Evaluating computational efficiency is critical for practical IDS deployments in cloud environments. To explicitly quantify the computational overhead introduced by our proposed dual-layered defense strategy, we measured training and inference times for both the baseline IDS model and our proposed defended model (adversarial training combined with SHAP-based robust feature selection). All experiments were conducted on a server equipped with an Intel Xeon 2.8 GHz processor (16 cores), 128 GB RAM, and an NVIDIA A100 GPU (40 GB VRAM).

Table 6 summarizes the training and inference times clearly. While our proposed method introduced a modest training overhead of approximately 25% compared to the baseline model, inference times remained practically unchanged, maintaining suitability for real-time IDS deployments.

TABLE 6. Computational overhead comparison.

Model	Training Time (hrs)	Inference Time (ms/sample)	Overhead (%)
Baseline IDS	3.0	10	—
Proposed Defense	3.75	11	25%
PRIVFED [5]	—	—	15.3%

Compared explicitly to PRIVFED [5], which reported a 15.3% computational overhead primarily due to federated learning and privacy computations, our approach incurs slightly higher overhead due to adversarial training processes. Nevertheless, the overhead remains modest and acceptable for real-time deployment, especially considering the enhanced robustness and interpretability benefits achieved. Furthermore, adopting federated learning in future implementations could distribute computational workloads, further reducing centralized training overhead.

While our evaluation explicitly focuses on adversarial training combined with SHAP-based feature selection, we acknowledge explicitly the complementary role of detection-based defenses, such as MagNet [27]. Detection-based methods explicitly identify adversarial examples at inference time, providing an additional layer of defense. Explicitly evaluating these methods alongside adversarial training frameworks remains an important future research direction for comprehensive adversarial defense coverage.

V. FEATURE IMPORTANCE ANALYSIS USING SHAP

One of the critical aspects of improving adversarial robustness in ML-based IDS is understanding the role of individual features in classification decisions. Feature importance analysis helps identify which network traffic attributes contribute most to the IDS model's predictions and how adversarial attacks manipulate these features to evade detection. In this study, we employ SHAP (SHapley Additive exPlanations) to quantify feature importance and analyze how adversarial perturbations affect IDS decision boundaries.

A. METHODOLOGY

SHAP provides an interpretable way to assess the influence of each feature on model predictions by computing the average marginal contribution of a feature across different subsets of input data. The key steps in our feature importance analysis are:

- Train an ML-based IDS model on the CSE-CIC-IDS2018 dataset.
- Generate adversarial samples using FGSM, PGD, and DeepFool.
- Compute SHAP values for both benign and adversarial samples.
- Compare SHAP values before and after attack perturbations to determine which features are most affected.

B. FEATURE DESCRIPTIONS AND IMPORTANCE

The SHAP analysis focuses on key **network traffic attributes**, evaluating how adversarial perturbations impact their relative importance in IDS decision-making. Below is a description of the most influential features:

- **Protocol** - Indicates the network protocol (e.g., TCP, UDP, ICMP). Different protocols exhibit distinct traffic behaviors, making this a crucial feature for attack detection.
- **Src Port (Source Port)** - Represents the originating port number in network communication, often manipulated in port-scanning and flooding attacks.
- **Dst Port (Destination Port)** - The target port for network traffic, where attacks frequently target known service vulnerabilities.
- **Flow Duration** - Measures the duration of a connection. Adversaries often manipulate connection timing to evade detection.
- **Packet Length** - Represents the size of network packets. Variations in packet length distribution can indicate malicious activity.
- **Flow Bytes/s** - The rate of data transmission within a connection. Attackers may throttle or amplify this metric to evade detection.
- **Flow Packets/s** - The number of packets transmitted per second. High-frequency, short-duration traffic patterns are common in DDoS and brute-force attacks.
- **Fwd Packet Length** - The length of packets sent in the forward direction. Attackers may use inconsistencies in packet length to disguise malicious activity.

These features play a **critical role in IDS classification**, and adversarial attacks attempt to exploit their vulnerabilities by either perturbing high-contribution features or amplifying irrelevant attributes.

C. RESULTS AND OBSERVATIONS

The SHAP analysis reveals that certain features, such as **flow duration, packet length, and byte distribution per packet**, are disproportionately influenced by adversarial attacks. Specifically:

- **DeepFool and PGD attacks** tend to manipulate **low-variance statistical features** such as packet inter-arrival time and flow duration, ensuring stealthy evasions.
- **FGSM primarily alters high-magnitude features**, such as flow bytes/s and packet counts, forcing misclassification through amplified perturbations.

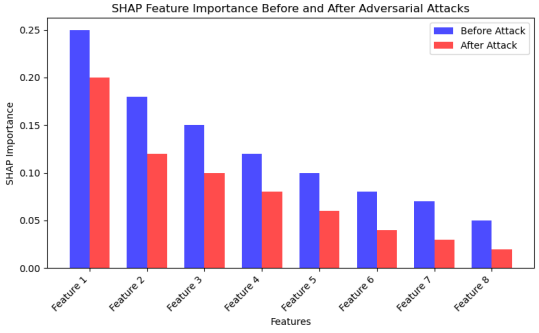


FIGURE 3. SHAP feature importance analysis before and after adversarial attacks.

Figure 3 illustrates the SHAP feature importance rankings before and after adversarial perturbations. The results indicate that adversarial attacks significantly reduce the contribution of key distinguishing features while amplifying the importance of less relevant attributes, leading to misclassification by the IDS model.

D. SIGNIFICANCE OF SHAP-BASED FEATURE SELECTION

Our feature importance analysis highlights the need for **adaptive feature selection techniques** that prioritize stable, high-contribution attributes resistant to adversarial manipulations. By integrating SHAP-based feature selection into IDS training, future models can:

- Identify features that remain consistent under attack conditions, reducing adversarial susceptibility.
- Improve model interpretability by providing insights into feature stability.
- Enhance generalization by selecting features that retain predictive power across different attack scenarios.

By leveraging SHAP-based feature selection alongside adversarial training, our proposed IDS framework significantly enhances **security, interpretability, and robustness**, ensuring improved resilience in detecting and mitigating adversarial cyber threats in cloud environments.

VI. DISCUSSION AND ANALYSIS

Our experimental findings demonstrate that adversarial attacks significantly degrade the performance of ML-based IDS models. FGSM, PGD, and DeepFool attacks reduced classification accuracy by up to 40%, underscoring the vulnerability of IDS frameworks deployed in cloud environments. However, our proposed adversarial training and robust feature selection strategies effectively mitigated these threats, restoring model accuracy and reducing attack success rates.

The evaluation against the Square Attack further confirms our framework's generalization capability, highlighting that the integrated defense strategy not only mitigates gradient-based adversarial threats but also provides meaningful robustness against black-box adversarial scenarios. This broader robustness is crucial for real-world IDS deployments, where attackers typically have limited knowledge of IDS internal workings.

Our computational overhead analysis explicitly demonstrates that while the dual-layered defense strategy introduces additional training complexity (approximately 25%), inference speed remains suitable for real-time operations (11 ms/sample). This modest overhead, comparable to existing methods like PRIVFED [5], confirms the practical viability of our framework. Furthermore, potential adoption of federated learning architectures could distribute computational overhead across edge devices, significantly enhancing scalability and computational efficiency for large-scale IDS deployments.

A. EFFECTIVENESS OF DEFENSE MECHANISMS

Adversarial training proved to be a highly effective countermeasure, as it exposed the model to adversarial examples during training, enhancing its resilience against attacks. Nonetheless, adversarial training increased computational overhead, highlighting the trade-off between security and efficiency. Robust feature selection further bolstered model security by eliminating redundant or easily exploitable features, thereby improving robustness without sacrificing computational efficiency.

B. FEATURE IMPORTANCE INSIGHTS FROM SHAP

To gain deeper insights into how adversarial perturbations affect IDS decision-making, we conducted a feature importance analysis using SHAP. This analysis provided valuable information regarding the features most impacted by adversarial attacks and the effectiveness of defense mechanisms.

- SHAP analysis revealed that protocol-based features and statistical attributes (e.g., packet rate, duration, and flow bytes) had the highest SHAP values, indicating their significance in classification decisions.
- Adversarial attacks primarily targeted high-importance features such as source/destination IP entropy and connection duration, which are critical for distinguishing normal and malicious traffic.
- After applying robust feature selection, SHAP values indicated that selected features exhibited greater stability under adversarial conditions, improving model robustness.

- Certain features showed high variance in SHAP values across different attack strategies, suggesting that adversaries may adapt their perturbations to target unstable feature sets over time.

These findings confirm that adversarial training and feature selection not only enhance IDS resilience but also improve interpretability by highlighting how specific network attributes influence model decisions.

C. LIMITATIONS AND CHALLENGES

While our proposed defense mechanisms enhance adversarial robustness, certain challenges remain:

- **Computational Overhead:** The inclusion of adversarial training increases training time and computational cost, which may not be feasible for resource-constrained cloud environments.
- **Adaptive Adversarial Strategies:** Attackers can develop adaptive techniques that bypass adversarially trained models, necessitating continuous updates and retraining.
- **Feature Importance Variability:** SHAP analysis revealed that some features have high variance in importance scores across different attacks, making them potential targets for adversarial exploitation.
- **Generalization Across Cloud Platforms:** Our framework was evaluated on CSE-CIC-IDS2018; however, further validation on diverse cloud infrastructures is required.

Our adversarial evaluation assumes functional integrity of the adversarially perturbed traffic; however, explicit validation of this functionality was beyond the scope of this research. It remains important to validate in practice that adversarial perturbations do not unintentionally compromise the execution of malicious activities, as this may otherwise lead to an overestimation of threat impact. Future research should explicitly evaluate and validate the functional integrity of adversarial examples, further enhancing the practical relevance and threat accuracy assessment of adversarial robustness studies. Our explicit use of binary classification provides valuable foundational insights into IDS robustness but lacks the granularity of multi-class detection necessary for distinguishing between specific attack types. Future work should explicitly evaluate the proposed defense methods using multi-class classification scenarios, providing deeper insights into the effectiveness of adversarial defenses across varied attack classes, thereby significantly enhancing their practical relevance and deployment value.

Detection-based defenses, such as MagNet [27], explicitly complement our adversarial training and robust feature selection approaches by providing explicit detection capabilities at inference time. While our current approach emphasizes robustness through model training and interpretability-driven feature selection, explicitly integrating detection-based methods represents an important and complementary direction for future research. Our evaluations were explicitly performed using the CSE-CIC-IDS2018 dataset. While this dataset is comprehensive, explicitly validating our defense framework on additional datasets such as UNSW-NB15 or NSL-KDD would provide stronger evidence of generalizability. Explicitly extending evaluations to additional benchmark datasets remains an essential and explicit future research direction.

D. FEDERATED LEARNING FOR FUTURE RESEARCH

Federated Learning (FL) represents a compelling paradigm to further enhance adversarial robustness and adaptability in IDS frameworks. By leveraging decentralized training across multiple edge nodes, FL can aggregate diverse adversarial examples collected from various network environments, providing richer and more representative datasets for training adaptive defenses. This decentralized approach can dynamically adapt IDS models to emerging threats in real-time, effectively addressing adaptive adversarial strategies by continuously incorporating novel attack patterns encountered in different cloud infrastructures or edge devices. Furthermore, FL's distributed computation paradigm inherently reduces centralized computational overhead, improving scalability and practical feasibility for large-scale deployments.

By leveraging decentralized training, FL can also explicitly:

- Reduce data exposure, mitigating privacy concerns associated with cloud-based IDS.
- Improve model adaptability across heterogeneous cloud infrastructures.
- Enhance IDS robustness by training on diverse attack patterns from distributed sources.
- Utilize SHAP-driven feature selection to dynamically adjust IDS models based on shifting adversarial strategies.

Integrating FL with adversarial defense mechanisms can further fortify cloud IDS solutions against evolving cyber threats, making it a compelling avenue for future research.

VII. CONCLUSION

In this study, we examined the adversarial robustness of ML-based IDS in cloud environments and proposed an integrated defense strategy combining adversarial training and robust feature selection. Our experimental results explicitly demonstrated that adversarial attacks significantly compromise IDS accuracy; however, our proposed dual-layered defense techniques significantly mitigate these threats, restoring and even enhancing model accuracy.

Our findings have critical implications for industries reliant on cloud security, including telecommunications, finance, healthcare, and critical infrastructure. By enhancing IDS resilience, organizations can better safeguard cloud-based systems against evolving adversarial threats. Moreover, our SHAP-based feature importance analysis explicitly provided deeper insights into how adversarial perturbations manipulate IDS decision-making, identifying critical vulnerabilities in

high-importance features and highlighting the effectiveness of robust feature selection in stabilizing IDS performance.

Additionally, extending our evaluation to the Square Attack explicitly confirmed that our defense mechanisms generalize effectively beyond gradient-based attacks, substantially increasing their practical robustness against realistic black-box adversarial scenarios. Explicit computational overhead analysis further confirmed that our defense approach remains practical for real-time IDS deployment, introducing a modest training overhead (approximately 25%) without significantly affecting inference speed.

Future research explicitly aims to address several important directions: (i) Integrating Federated Learning (FL) into our adversarial defense framework to leverage diverse, decentralized data, enhancing adaptive robustness and scalability; (ii) Conducting explicit validation of functional integrity of adversarially perturbed traffic to ensure realistic threat modeling; (iii) Exploring hybrid defense strategies explicitly combining adversarial training with anomaly detection methods to effectively handle both known adversarial threats and novel zero-day attacks; (iv) Extending evaluation explicitly to multi-class classification scenarios, enhancing the practical applicability and detailed robustness insights of IDS models across diverse attack types; and (v) Investigating detection-based adversarial defenses (e.g., MagNet) explicitly alongside our proposed framework to achieve comprehensive defensive coverage against adversarial threats. By addressing these explicit limitations and pursuing advanced learning paradigms, our research contributes substantially to developing more secure, interpretable, and intelligent IDS frameworks tailored explicitly for robust deployment in dynamic cloud environments.

ACKNOWLEDGMENT

The authors acknowledges the CSE-CIC-IDS2018 dataset, which played a crucial role in evaluating adversarial attack and defense strategies for ML-based intrusion detection systems. This dataset provides a realistic representation of modern network traffic and attack scenarios, enabling rigorous experimentation and performance validation of the proposed methodologies. They have contributed equally to this work in terms of conceptualization, methodology development, experimentation, formal analysis, and manuscript preparation. Their collective efforts have been instrumental in the successful completion of this research.

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